

Brasília, Brazil

Dear Editors,

Please consider our manuscript, “*Trustworthy Selective Computation for Resource-Constrained Vehicular Edge Networks*”, for publication in Ad Hoc Networks, within the special issue *VSI: Trustworthy and Resilient Edge AI Systems*.

The paper studies when selective computation—skipping or reusing inference steps to stay within energy and latency budgets—can be trusted in strictly causal vehicular edge inference. We show, through a controlled factorial study with a deterministic risk onset, that the safety of a runtime scheduler depends on the temporal organization of the model’s predictive uncertainty rather than on the scheduling rule or probabilistic calibration: an entropy-gated scheduler applied to unorganized trajectories worsens the effective detection delay by 11.8% without reducing misses, while the same scheduler on trajectories organized by our offline induction protocol (AF-TOI) achieves 53.4% effective cost reduction with coverage preserved. The deployed pipeline fits within 58K parameters and 1 KB of recurrent state, and emits a compact operational-risk signal suitable as a coordination primitive for vehicular edge architectures, which we believe fits the scope of the special issue directly.

This manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors have approved the manuscript and agree with its submission. The authors declare no competing interests.

Thank you for your consideration.

Sincerely,

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